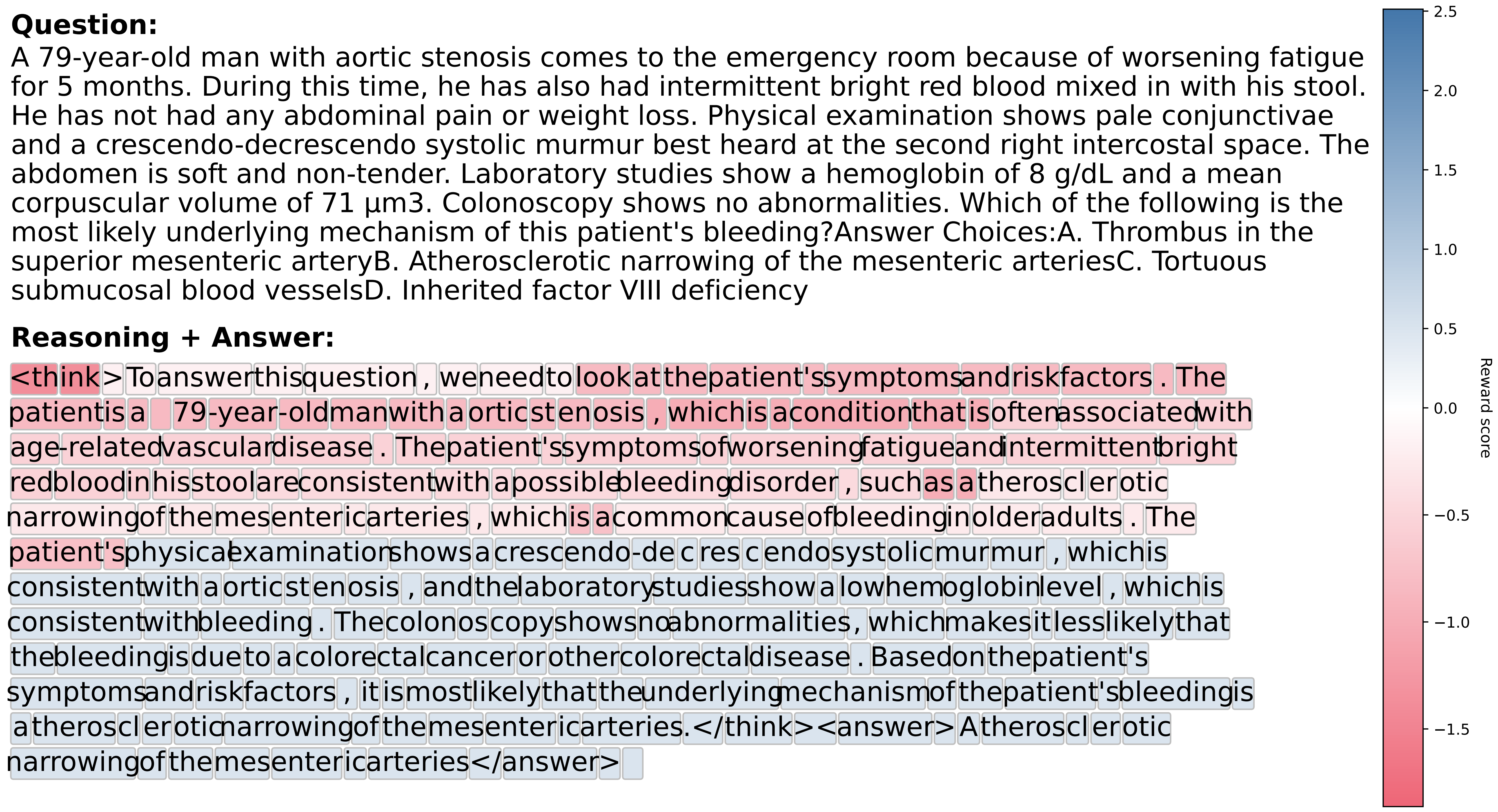
Question:
A 79-year-old man with aortic stenosis comes to the emergency room because of worsening fatigue for 5 months. During this time, he has also had intermittent bright red blood mixed in with his stool. He has not had any abdominal pain or weight loss. Physical examination shows pale conjunctivae and a crescendo-decrescendo systolic murmur best heard at the second right intercostal space. The abdomen is soft and non-tender. Laboratory studies show a hemoglobin of 8 g/dL and a mean corpuscular volume of 71 μm3. Colonoscopy shows no abnormalities. Which of the following is the most likely underlying mechanism of this patient's bleeding?Answer Choices:A. Thrombus in the superior mesenteric arteryB. Atherosclerotic narrowing of the mesenteric arteriesC. Tortuous submucosal blood vesselsD. Inherited factor VIII deficiency
Reasoning + Answer:
<think> To answer this question, we need to look at the patient's symptoms and risk factors. The patient is a 79-year-old man with aortic stenosis, which is a condition that is often associated with age-related vascular disease. The patient's symptoms of worsening fatigue and intermittent bright red blood in his stool are consistent with a possible bleeding disorder, such as atherosclerotic narrowing of the mesenteric arteries, which is a common cause of bleeding in older adults. The patient's physical examination shows a crescendo-decrescendo systolic murmur, which is consistent with aortic stenosis, and the laboratory studies show a low hemoglobin level, which is consistent with bleeding. The colonoscopy shows no abnormalities, which makes it less likely that the bleeding is due to a colorectal cancer or other colorectal disease. Based on the patient's symptoms and risk factors, it is most likely that the underlying mechanism of the patient's bleeding is atherosclerotic narrowing of the mesenteric arteries.</think><answer>Atherosclerotic narrowing of the mesenteric arteries</answer>
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Reward score